

7 Demonstration of Multithreading using Thread Class and Runnable Interface

```
i. class S1 extends Thread{
    S1()
    {
        System.out.println("starting thread1");
    }
    public void run(){
        System.out.println("task one");
    }
}
```

```
class S2 extends Thread
{
    S2()
    {
        System.out.println("starting thread2");
    }
}
```

```
public void run(){
    System.out.println("task two");
}
}
```

```
class multithread{
    public static void main(String args[]){
        S1 t1=new S1();
        S2 t2=new S2();
        t1.start();
        t2.start();

    }
}
```

Output:

```
C:\Users\reva00381\Desktop\Java>java multithread
starting thread1
starting thread2
task one
task two
```

ii. Class NewThread implements Runnable {

```
Thread t;
NewThread() {
    // Create a new, second thread
    t = new Thread(this, "Demo Thread");
    System.out.println("Child thread: " + t);
    t.start(); // Start the thread
}
```

```

// This is the entry point for the second thread.
public void run() {
    try {
        for(int i = 5; i > 0; i--) {
            System.out.println("Child Thread: " + i);
            Thread.sleep(500);
        }
    } catch (InterruptedException e) {
        System.out.println("Child interrupted.");
    }
    System.out.println("Exiting child thread.");
}

class ThreadDemo {
    public static void main(String args[]) {
        new NewThread(); // create a new thread
        try {
            for(int i = 5; i > 0; i--) {
                System.out.println("Main Thread: " + i);
                Thread.sleep(1000);
            }
        } catch (InterruptedException e) {
            System.out.println("Main thread interrupted.");
        }
        System.out.println("Main thread exiting.");
    }
}

```

## 8 Demonstration of GUI using Swings

```

import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class EvenOddChecker extends JFrame {

    private JTextField numberInput;
    private JButton checkButton;
    private JLabel resultLabel;

    public EvenOddChecker() {
        // Set up the JFrame
        setTitle("Even or Odd Checker");
        setSize(300, 200);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        // Create components
    }
}

```

```

numberInput = new JTextField(10);
checkButton = new JButton("Check");
resultLabel = new JLabel("Result will be displayed here");

// Set up layout
JPanel panel = new JPanel();
panel.add(new JLabel("Enter a number:"));
panel.add(numberInput);
panel.add(checkButton);
panel.add(resultLabel);

// Add action listener for the button
checkButton.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        checkEvenOdd();
    }
});

// Add panel to frame
add(panel);
}

private void checkEvenOdd() {
    try {
        // Get the number from the text field
        int number = Integer.parseInt(numberInput.getText());
        // Check if the number is even or odd
        if (number % 2 == 0) {
            resultLabel.setText(number + " is even.");
        } else {
            resultLabel.setText(number + " is odd.");
        }
    } catch (NumberFormatException e) {
        resultLabel.setText("Please enter a valid integer.");
    }
}

public static void main(String[] args) {
    // Run the application
    SwingUtilities.invokeLater(() -> {
        EvenOddChecker checker = new EvenOddChecker();
        checker.setVisible(true);
    });
}
}

```